

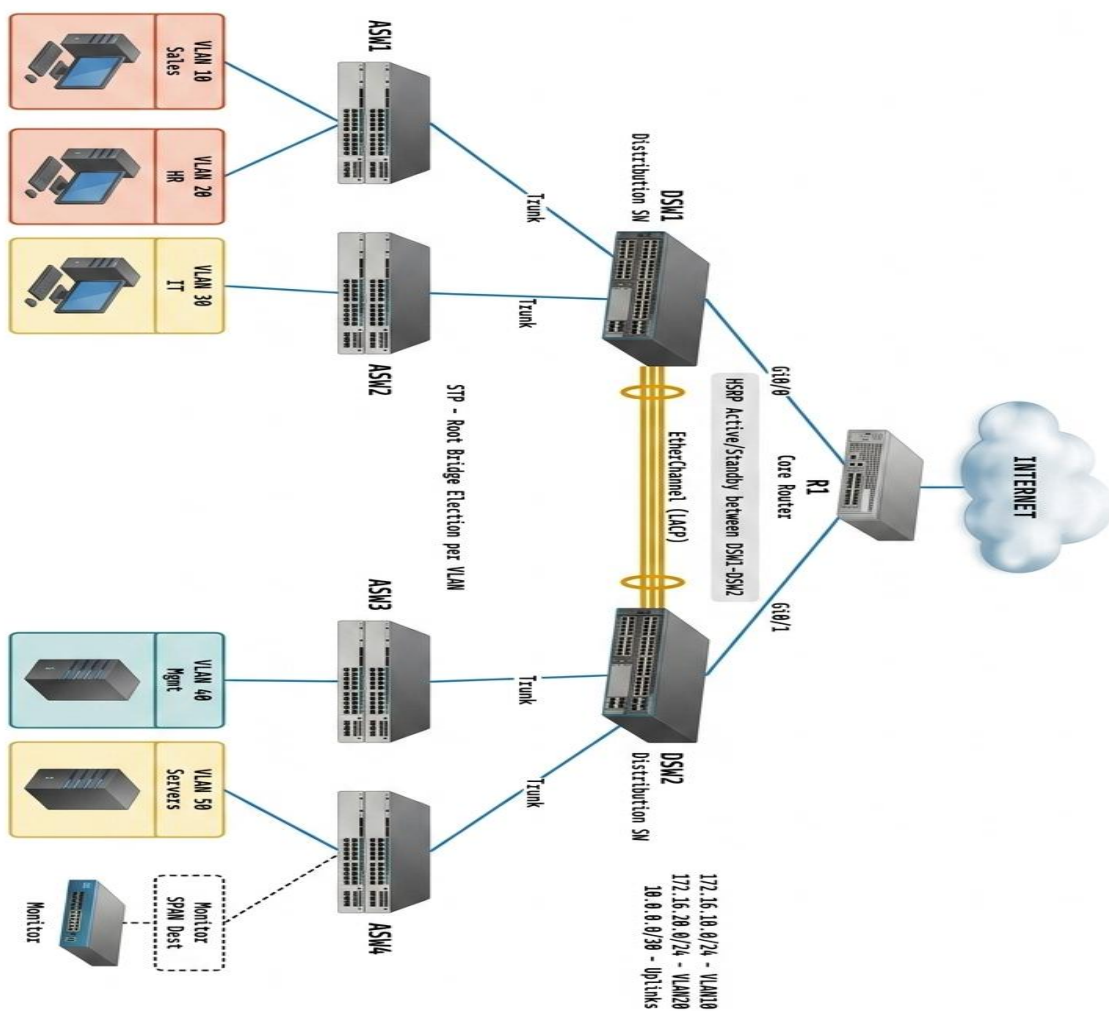


National youth Services Council
Diploma in Computer Networking
&
Cyber Security
2026 January – June
Final Examination



Answer all the questions.

Time: - 2 hours



QUESTION 01: VLAN CONFIGURATION & PORT SECURITY (25 MARKS)

Using the topology above, configure VLANs and implement port security on ASW1 and ASW2 in Packet Tracer/GNS3.

- A** Create VLANs 10 (Sales), 20 (HR), 30 (IT), 40 (Mgmt) and 99 (Native) on all switches. Assign names to each VLAN. (6 marks)
- B** Configure trunk ports between DSW1↔ASW1, DSW1↔ASW2 allowing only VLANs 10, 20, 30, 40 with VLAN 99 as native. (6 marks)
- C** Configure access ports on ASW1: Fa0/1-10 → VLAN 10, Fa0/11-20 → VLAN 20. Enable Port Security: max 2 MACs, violation shutdown. (8 marks)
- D** Verify using show vlan brief, show interfaces trunk, and show port-security interface. Paste output. (5 marks)

» Note: Sticky MAC learning must be enabled on all secured ports. Err-disabled recovery must be configured for violation recovery.

QUESTION 02: ETHERCHANNEL & SPANNING TREE PROTOCOL (STP) (25 MARKS)

Configure Layer 2 EtherChannel between DSW1 and DSW2 using LACP, and tune STP to control root bridge election per VLAN.

- A** Configure EtherChannel using LACP (802.3ad) on interfaces Gi1/0–Gi1/1 between DSW1 and DSW2. Assign to Port-Channel 1, configure as trunk. (7 marks)
- B** Configure Rapid PVST+. Set DSW1 as primary root for VLANs 10 and 20. Set DSW2 as primary root for VLANs 30 and 40. (8 marks)
- C** Enable PortFast and BPDU Guard on all access ports of ASW1 and ASW2. (5 marks)
- D** Verify via show etherchannel summary, show spanning-tree vlan 10. Confirm root bridge roles and port states. (5 marks)

QUESTION 03: OSPF ROUTING & SPAN (PORT MIRRORING) (25 MARKS)

Configure OSPF Area 0 between R1, DSW1, and DSW2 for inter-VLAN routing. Set up SPAN on ASW4 for traffic monitoring.

- A** Configure SVIs on DSW1 and DSW2 for all VLANs with IP helper addresses. Enable IP routing on both distribution switches. (7 marks)
- B** Configure OSPF Area 0 on R1, DSW1, DSW2. Advertise all VLAN subnets. Set DSW1 as DR with priority 200. (8 marks)

- C** Configure SPAN session 1 on ASW4: monitor source VLAN 50 (Servers) → destination port Fa0/24 (Monitor PC). (5 marks)
- D** Verify OSPF neighbors (show ip ospf neighbor), routing table (show ip route ospf), and SPAN (show monitor session 1). (5 marks)

QUESTION 04: HSRP CONFIGURATION & STATIC/DEFAULT ROUTING (25 MARKS)

Implement HSRP between DSW1 and DSW2 for gateway redundancy, and configure static routing on R1 for internet connectivity.

- A** Configure HSRP Group 10 on VLAN 10 SVI: DSW1 = Active (priority 110), DSW2 = Standby. Virtual IP = 172.16.10.254. Enable preempt. (8 marks)
- B** Configure HSRP Group 20 on VLAN 20: DSW2 = Active, DSW1 = Standby. Demonstrate load balancing per VLAN. (6 marks)
- C** Configure static default route on R1 pointing to the ISP. Redistribute into OSPF using default-information originate always. (6 marks)
- D** Test failover by shutting DSW1's uplink. Verify HSRP role change (show standby brief) and continued ping to gateway. (5 marks)